

Notes: Howell (AER, 2017)

Financing Innovation: Evidence from R&D Grants

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Relative to prior R&D subsidy studies	3
2.2	Relative to entrepreneurial finance and venture capital research	3
2.3	Relative to the certification literature	4
2.4	Relative to innovation policy	4
3	Data and measurement	4
3.1	The DOE SBIR program	4
3.2	Application data	5
3.3	Outcome variables	5
3.4	Technology and financing-constraint measures	5
4	Empirical strategy and specifications	6
4.1	Sharp regression discontinuity around the award cutoff	6
4.2	Negative binomial and OLS patent specifications	6
4.3	VC, survival, and exit specifications	6
4.4	Revenue specifications	6
4.5	Phase 2 specification	7
4.6	Heterogeneity specification	7
5	Identification and validity	7
5.1	The core RD assumption	7
5.2	Balance and manipulation tests	7
5.3	Certification versus funding	8
5.4	Potential limitations	8
6	Tables and figures	8
6.1	Table 1 and Figure 1: sample and RD balance	8
6.2	Figure 2 and Table 2: Phase 1 grant impact on patents	9
6.3	Figure 3 and Table 3: Phase 1 grant impact on venture capital	10
6.4	Figure 4 and Table 4: Revenue effects	11
6.5	Table 5 and Table 6: Survival, exit, and Phase 2 effects	12
6.6	Table 7 and Table 8: Heterogeneity and technology categories	13
6.7	Figure 5 and Table 9: Timing and grant use	13
6.8	Table 10 and Table 11: Robustness and placebo tests	14
7	Short summary	14
8	Conclusion	15
8.1	Core conclusion	15
8.2	Economic intuition	15

8.3	Incremental contribution in one sentence	15
8.4	Takeaway	15

1 Big picture

- **Question.** Do early-stage government R&D grants help innovative startups, or do they crowd out private investment and fund projects that would have happened anyway?
- **Setting.** The paper studies the U.S. Department of Energy’s Small Business Innovation Research (SBIR) grant program. Phase 1 grants are small early-stage grants, historically around \$150,000, intended to support feasibility testing and proof-of-concept R&D. Phase 2 grants are much larger, roughly \$1 million, and occur later.
- **Main empirical design.** DOE program officials rank applicants within sector-specific competitions. The number of awards in a competition creates an award cutoff. The paper uses a sharp regression discontinuity design comparing firms just above and just below the cutoff.
- **Main result.** Phase 1 grants have large positive effects on innovation, private financing, commercialization, survival, and successful exit.
- **Patent result.** A Phase 1 award increases cite-weighted patenting. In the negative binomial specification, awardees produce about 2.5 times as many cite-weighted patents as comparable nonawardees.
- **Financing result.** A Phase 1 award nearly doubles the probability of receiving subsequent venture capital, from roughly 10 percent to 19 percent. It also increases the amount of VC raised and the number of deals.
- **Revenue result.** A Phase 1 award increases revenue. In the main revenue specifications, it raises log revenue or has a large positive effect on revenue conditional on positive sales.
- **Survival and exit result.** A Phase 1 award increases the probability that a firm remains in business and increases the probability of a successful exit through IPO or acquisition.
- **Phase 2 contrast.** Phase 2 grants have little measurable effect on outcomes other than a small effect on patents. This is partly because many high-quality Phase 1 winners that raise VC do not apply for Phase 2.
- **Mechanism.** The evidence favors a funding/prototyping channel over a pure certification channel. The grant helps firms demonstrate technology feasibility and reduce early-stage uncertainty.
- **Policy takeaway.** Carefully targeted early-stage R&D subsidies can relax financing frictions and catalyze private investment rather than crowding it out.

2 Incremental contribution of the paper

2.1 Relative to prior R&D subsidy studies

- Earlier work on public R&D subsidies often faced a selection problem: grant recipients are more promising than nonrecipients, so simple comparisons do not identify causal effects.
- Howell contributes the first large-sample quasi-experimental evaluation of R&D grants to private firms using internal ranked application data.
- The paper directly implements the regression discontinuity strategy proposed in the literature for grant evaluation: compare applicants just above and just below the funding cutoff.
- The sample is unusually broad: thousands of private high-tech firms and applications across many DOE technology areas over a long period.

Interpretation: The incremental empirical value is the combination of proprietary ranking data and rich startup outcomes. This allows the paper to move from “grant recipients do well” to “grants causally change outcomes for marginal applicants.”

2.2 Relative to entrepreneurial finance and venture capital research

- The paper directly links public grants to subsequent private financing, especially venture capital.
- It shows that early-stage grants can make some firms investable rather than substituting for venture capital.

- It provides direct evidence that financial constraints are important for young innovative firms.
- The heterogeneity results show stronger effects for firms that plausibly face more severe constraints: younger firms, less experienced firms, hardware firms, and firms in emerging sectors.

Interpretation: The paper adds a causal test to a central entrepreneurial finance claim: external finance frictions prevent innovative startups from making value-enhancing investments.

2.3 Relative to the certification literature

- A grant could help startups by certifying quality to outside investors.
- Howell designs a direct test: if certification is the main channel, DOE ranks should convey information about future success because ranks are the latent quality scores behind award status.
- The paper finds that ranks are mostly uninformative for market outcomes once award status is controlled for.
- The grant amount matters, and survey evidence indicates that firms use Phase 1 funding for R&D, testing, and prototyping rather than only as a signal.

Interpretation: The paper’s incremental contribution is to separate *being selected by the government* from *receiving money to build a prototype*. The evidence points toward the latter.

2.4 Relative to innovation policy

- The paper shows that small, early grants can have large downstream effects on private financing and commercialization.
- It also shows that larger later-stage grants may have much weaker marginal effects when higher-quality firms have already attracted private capital.
- This distinction matters for policy design: public money is likely most useful at the proof-of-concept stage, where private investors face high uncertainty and startups have limited collateral.

Interpretation: The paper shifts the innovation-policy question from “Should governments subsidize R&D?” to “At what stage and for which firms are subsidies most effective?”

3 Data and measurement

3.1 The DOE SBIR program

- The SBIR program is a federal program that funds R&D by small businesses.
- The DOE program has two main stages.
- **Phase 1** provides early-stage proof-of-concept grants. The award is small enough to be unlikely to finance full commercialization but large enough to fund feasibility work.
- **Phase 2** provides larger follow-on funding after Phase 1. Phase 1 winners are eligible to apply for Phase 2 roughly nine months later.
- DOE competitions are sector-specific. Each competition has a ranked list of applicants and a cutoff determined by budget and award availability.

Interpretation: The rank and cutoff structure creates the paper’s research design. Firms close to the cutoff are similar in DOE-evaluated quality, but one side receives funding and the other does not.

3.2 Application data

- The proprietary DOE dataset contains applications, firm identities, funding status, award dates, ranking information, and competition identifiers.
- The broader data cover 14,522 Phase 1 applications and 7,419 unique Phase 1 applicant firms from 1983 to 2013.
- The regression discontinuity sample uses ranked Phase 1 applications from 1995 to 2013.
- The main RD sample contains 5,021 ranked Phase 1 applications and 428 competitions.
- Phase 2 analysis uses the subset of Phase 1 winners that apply for Phase 2.

Interpretation: The key data advantage is that the paper observes rejected applicants and their rankings. Most grant evaluation studies observe only winners or lack the information needed to compare winners with close losers.

3.3 Outcome variables

The paper tracks post-award firm performance through several outcome categories.

- **Innovation:** cite-weighted patents and patent counts, matched to patent application dates.
- **Private financing:** indicators for subsequent venture capital investment, amounts raised, and number of VC deals.
- **Commercialization:** revenue as of 2016 and log revenue.
- **Survival:** indicator for whether the firm remains in business.
- **Successful exit:** IPO or acquisition.
- **Firm characteristics:** age, previous patents, previous SBIR awards, women-owned status, technology type, local MSA venture capital activity, and MSA income.

Interpretation: The outcome set is important because a grant could affect scientific output, external financing, commercial revenue, or survival differently. The paper therefore tests the full path from R&D input to private market validation.

3.4 Technology and financing-constraint measures

- **Hardware versus software:** hardware firms often require more upfront capital and are harder to finance with intangible collateral.
- **Previous patents/citations:** firms with more previous patents may have more reputation, more collateralizable knowledge, and less uncertainty.
- **Age:** younger firms have shorter track records and are more likely to face financing constraints.
- **Emerging sector:** firms in newer sectors face greater investor uncertainty.
- **Technology type:** the paper splits effects by DOE technology categories such as geothermal, solar, fuel cells, and smart grid.

Interpretation: These variables help test whether the grant is most effective where private financing frictions should bind most strongly.

4 Empirical strategy and specifications

4.1 Sharp regression discontinuity around the award cutoff

Let i index firms and c index competitions. Let $Rank_{ic}$ be the firm's rank normalized around the cutoff, where the lowest-ranked winner has positive rank and the highest-ranked loser is just below zero. Let $Award_{ic}$ be an indicator equal to one if firm i wins the Phase 1 grant.

The baseline specification is:

$$Y_{ic}^{post} = \alpha + \beta Award_{ic} + f(Rank_{ic}) + \delta_c + \Gamma' X_{ic} + \varepsilon_{ic}. \quad (1)$$

- Y_{ic}^{post} is a post-award outcome such as cite-weighted patents, VC, revenue, survival, or exit.
- $Award_{ic}$ is the treatment.
- $f(Rank_{ic})$ is a polynomial or rank control.
- δ_c are competition fixed effects when feasible.
- X_{ic} includes baseline controls in some specifications, including pre-award outcome measures and previous non-DOE SBIR awards.

Interpretation: The coefficient β compares winners and losers near the cutoff. The identifying assumption is that, conditional on smooth functions of rank, firms immediately above and below the cutoff are comparable.

4.2 Negative binomial and OLS patent specifications

For cite-weighted patents, the paper estimates both count and linearized models:

$$E[Cites_{ic}^{post} | X] = \exp(\alpha + \beta Award_{ic} + f(Rank_{ic}) + \delta_c + \Gamma' X_{ic}), \quad (2)$$

for the negative binomial specification, and

$$\ln(1 + Cites_{ic}^{post}) = \alpha + \beta Award_{ic} + f(Rank_{ic}) + \delta_c + \Gamma' X_{ic} + \varepsilon_{ic} \quad (3)$$

for OLS.

Interpretation: The count model preserves the patent count structure; the log transformation makes the effect easier to interpret as an approximate percentage change.

4.3 VC, survival, and exit specifications

For binary outcomes, the paper primarily uses OLS linear probability models:

$$Pr(Y_{ic}^{post} = 1) = \alpha + \beta Award_{ic} + f(Rank_{ic}) + \delta_c + \Gamma' X_{ic} + \varepsilon_{ic}. \quad (4)$$

- OLS is preferred because logit drops fixed-effect groups with no successes.
- VC outcomes include an indicator for receiving VC, amount of VC, and number of deals.
- Survival and exit are also modeled as probability outcomes.

Interpretation: The linear probability model preserves the RD comparison across competitions and gives coefficients in percentage points.

4.4 Revenue specifications

Revenue is skewed and has many zeros, so the paper uses several models:

- OLS for revenue in millions of dollars.
- Tobit models treating zeros as censored.
- Zero-inflated negative binomial models to separate the probability of zero revenue from the positive revenue distribution.

Interpretation: These models are useful because revenue combines two margins: whether the firm commercializes at all and how large revenue is conditional on commercialization.

4.5 Phase 2 specification

The Phase 2 analysis focuses on firms eligible for and applying to Phase 2. Because Phase 2 award decisions have too few applicants per competition for the same rank-control RD design, the paper uses sector-year controls:

$$Y_i^{post} = \alpha + \beta Phase2Award_i + \eta_{sector,year} + \Gamma' X_i + \varepsilon_i. \quad (5)$$

Interpretation: The Phase 2 result is less cleanly identified than Phase 1. It is still informative because it asks whether later, larger government funding has marginal effects after Phase 1.

4.6 Heterogeneity specification

To test financing constraints, the paper interacts the award with firm characteristics:

$$Y_{ic}^{post} = \alpha + \beta Award_{ic} + \lambda(Award_{ic} \times H_i) + \phi H_i + f(Rank_{ic}) + \delta_c + \varepsilon_{ic}, \quad (6)$$

where H_i is an indicator such as hardware firm, young firm, no previous patents, or emerging sector.

Interpretation: If the grant relaxes financing constraints, λ should be positive for firms that need more outside capital or have less investor-verifiable quality.

5 Identification and validity

5.1 The core RD assumption

- The RD design requires that potential outcomes evolve smoothly with rank around the cutoff.
- The grant award must be the only discontinuous change at the cutoff.
- Firms and DOE officials must not manipulate ranks around the cutoff.

Interpretation: Since the award cutoff is determined after ranking and the ranking rule is not known to firms, manipulation is difficult. Still, the paper tests for it carefully.

5.2 Balance and manipulation tests

The paper uses several validity checks:

- A density test checks that there is no discontinuity in the number of applicants around the cutoff.
- Figure 1 checks continuity in observables: age, future sector patenting, women-owned status, minority-owned status, previous SBIR awards, major-metro status, hardware status, MSA income, and MSA VC activity.
- Figure 2 panel A checks pre-award cite-weighted patents by rank.
- Figure 3 panel A checks pre-award VC by rank.
- Additional tests show that baseline covariates do not predict discontinuities in outcomes around the cutoff.

Interpretation: The validity tests support the idea that near-cutoff winners and losers are comparable before treatment.

5.3 Certification versus funding

- If grants work mainly by certification, DOE ranks should contain quality information that investors would value.
- The paper finds that ranks are relatively uninformative about future market outcomes once award status is controlled for.
- The grant effect varies with grant size and with financing-constraint proxies, supporting a funding/prototyping channel.
- Survey evidence shows that firms use Phase 1 funds for basic R&D, patent applications, and testing/demonstrating/prototyping.

Interpretation: Certification is not ruled out completely, but it cannot explain the full evidence. The main channel is that grant money allows firms to build and test technologies that reduce uncertainty for private investors.

5.4 Potential limitations

- The RD identifies local average treatment effects for firms near the award cutoff.
- The Phase 2 analysis is less cleanly identified than Phase 1.
- DOE SBIR firms are high-tech and energy-related; effects may not generalize to all startups or all grant programs.
- Revenue and survival measures are imperfect and may miss some forms of social value.
- The paper focuses on private outcomes, not the full social return to subsidized innovation.

Interpretation: The findings are strongest for marginal Phase 1 DOE SBIR applicants. The policy lesson is about early-stage R&D grants in high-uncertainty sectors, not unconditional endorsement of all subsidies.

6 Tables and figures

6.1 Table 1 and Figure 1: sample and RD balance

Read:

- Table 1 reports the main application, firm, and outcome summary statistics. The DOE application data include 14,522 Phase 1 applications and 7,419 unique applicant firms from 1983 to 2013.
- The RD sample from 1995 to 2013 includes 5,021 ranked Phase 1 applications and 428 competitions.
- Average Phase 1 competition size is 11.8 applicants, and the average number of awards per competition is 1.7.
- Baseline VC receipt is rare: only 8.3 percent have pre-award VC and 11 percent receive post-award VC.
- Figure 1 checks covariate balance around the cutoff for age, sector growth, ownership indicators, previous SBIR awards, metro status, hardware status, local income, and local VC activity.

Economic message: The table describes a sample of young, uncertain high-tech firms. The figure supports the RD design by showing that near-cutoff winners and losers look similar before treatment.

TABLE 1—SUMMARY STATISTICS

Panel A. Application data from DOE					
1983–2013					
Phase 1 applications					14,522
Unique Phase 1 applicant firms					7,419
Competitions					1,633
1995–2013					
Phase 1 applications					9,659
Unique Phase 1 applicant firms					4,545
Phase 1 applications with ranking data used in RD					5,021
Phase 1 competitions used in RD (≥ 1 award)					428
Average Phase 1 applicants per competition					11 (8.3)
Average Phase 1 awards per competition					1.7 (1.1)
Phase 2 applications used in RD					919
Panel B. Variables used in analysis from non-DOE sources					
	Type	Mean	SD	Median	Observations
Pre-award venture capital (VC) investment	0–1	0.083	0.27	0	5,021
Pre-award venture capital deals	Count	0.25	1.3	0	5,021
Pre-award cite-weighted patents	Count	21	122	0	5,021
Pre-award patents	Count	1.9	7.5	0	5,021
Pre-award acquisition or IPO	0–1	0.033	0.18	0	5,021
Post-award VC (VC_i^{post})	0–1	0.11	0.31	0	5,021
Post-award VC (mill 20125) ($VC_{Amt}_i^{post}$)	Cont.	2.7	26	0	4,964
Post-award VC deals ($VC_{Deals}_i^{post}$)	Count	0.32	1.4	0	5,021
Post-award cite-weighted patents ($Cites_i^{post}$)	Count	12	117	0	5,021
Post-award patents	Count	2	11	0	5,021
Post-award acquisition or IPO ($Exit_i^{post}$)	0–1	0.034	0.18	0	5,021
Revenue as of 2016 in \$ millions ($Revenue_i$)	Cont.	2.0	6.6	0.20	3,583
Survival as of 2016 ($IncstmBus_i^{surv}$)	0–1	0.67	0.47	1	3,880
Probability in major metro area (top 6)	0–1	0.30	0.46	0	5,021
Age (years)	Count	9.5	11	6	3,427
Probability tech is hardware ($Hardware_i$)	0–1	0.43	0.49	1	2,571
Probability new subsector ($EmergingSector_i$)	0–1	0.58	0.49	1	2,571
Probability minority owned	0–1	0.077	0.27	0	1,722
Probability woman owned	0–1	0.084	0.28	0	1,722
All-government SBIR wins ($SBIR_i$)	Count	10	36	0	5,021
Future patents in medal class	Count	9,758	11,809	5,453	1,583
MSA VC investment 2011 (\$millions)	Cont.	851	1,570	0	4,950
MSA median per cap. income 2011 (in \$thousands)	Cont.	56	14	56	4,603

Note: This table summarizes the DOE SBIR application data in panel A, and variables used in the regression analysis.

Table 1: Summary statistics

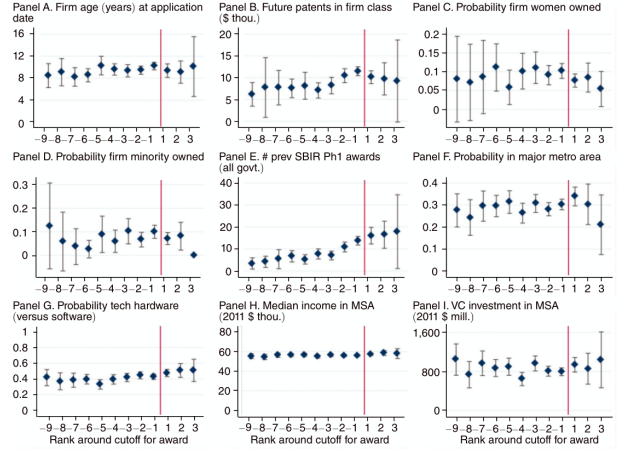


Figure 1: Covariate balance around cutoff

6.2 Figure 2 and Table 2: Phase 1 grant impact on patents

Read:

- Figure 2 shows no discontinuity in cite-weighted patents before the award decision.
- After the award decision, winners have much higher cite-weighted patenting.
- Table 2 shows that Phase 1 increases cite-weighted patents in both negative binomial and OLS specifications.
- In the main negative binomial specification for no previous winners, the award coefficient is about 0.91–0.93, implying approximately 2.5 times as many cite-weighted patents.
- OLS estimates show an increase of about 0.27–0.33 in log cite-weighted patents.

Economic message: The grant changes innovation outcomes, not just financing outcomes. It helps firms create patentable knowledge.

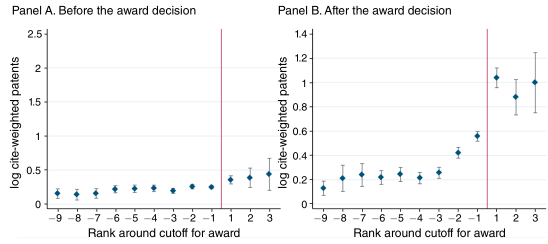


FIGURE 2. CITE-WEIGHTED PATENTS BEFORE AND AFTER PHASE 1 GRANT BY RANK

Figure 2: Cite-weighted patents by rank

TABLE 2—IMPACT OF PHASE 1 GRANT ON CITE-WEIGHTED PATENTS

Sample:	No previous winners					No prev.	>2 prev.
Bandwidth:	1	1	All	All	All	apps	wins
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A. Negative binomial model; dependent variable is $Cites_t^{post}$</i>							
Award	0.93 (0.21)	0.91 (0.19)	0.92 (0.33)	0.94 (0.26)	0.82 (0.13)	2.1 (0.34)	0.40 (0.14)
Normalized rank			0.052 (0.074)				
Normalized rank ²			−0.0072 (0.0045)				
Rank quintiles	No	No	No	Yes	No	No	No
Controls ^a	No	Yes	No	No	No	No	No
Sector-year fixed effects ^b	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,871	1,871	5,021	5,021	2,714	972	1,477
R ²	0.056	0.084	0.053	0.053	0.034	0.080	0.035
<i>Panel B. OLS models; dependent variable is $\ln(1 + Cites_t^{post})$</i>							
Award	0.33 (0.15)	0.27 (0.13)	0.29 (0.11)	0.22 (0.087)	0.29 (0.087)	0.49 (0.21)	0.22 (0.13)
Normalized rank			−0.026 (0.02)				
Normalized rank ²			7.8e-4 (0.0011)				
Rank quintiles	No	No	No	Yes	No	No	No
Controls ^d	No	Yes	No	No	No	No	No
Competition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,872	1,872	5,021	5,021	2,714	972	1,477
Pseudo-R ²	0.46	0.60	0.53	0.351	0.63	0.71	0.75

Notes: This table reports regression estimates of the Phase 1 award effect on cite-weighted patents using variants of equation (1). The model is negative binomial in panel A and OLS in panel B. 1–4 limit the sample to firms that have not previously won an award, but may have previously applied. 5–7 respectively use the whole sample, only firms that have never previously applied, and only firms that have more than two previous wins.

^a Controls are $Cites_t^{prev}$ or $\ln(1 + Cites_t^{prev})$ and previous non-DOE SBIR awards.

^b Competition fixed effects in the NB model do not permit convergence. Standard errors are clustered by sector-year. Year ≥ 1995 .

Table 2: Phase 1 grant impact on patents

6.3 Figure 3 and Table 3: Phase 1 grant impact on venture capital

Read:

- Figure 3 shows no discontinuity in VC before the award decision, but a clear discontinuity after the award decision.
- Table 3 shows that Phase 1 increases the probability of subsequent VC.
- The main VC probability effects are around 9–12 percentage points, relative to a baseline near 10 percent.
- The grant also increases log VC amount and the number of VC deals.
- The effect appears among firms with no previous winners and across broader samples.

Economic message: Phase 1 grants catalyze private capital. This is the opposite of pure crowd-out and is central evidence for the financing-constraint channel.

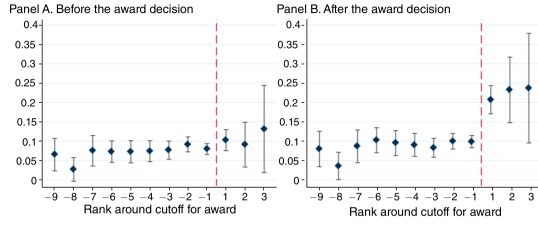


FIGURE 3. PROBABILITY OF VENTURE CAPITAL BEFORE AND AFTER GRANT BY RANK

Figure 3: VC probability by rank

Panel A							
Dependent variable:		VC_i^{post}					$\ln(1 + VC Amt_i^{post})$
Bandwidth:	1		3		All	1	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Award	0.098 (0.032)	0.10 (0.033)	0.12 (0.058)	0.094 (0.033)	0.072 (0.033)	0.10 (0.028)	1.40 (0.5)
Normalized rank			-0.029 (0.033)		0.0086 (0.0071)		
Normalized rank ²			0.012 (0.0088)		-7.4e-5 (4.3e-4)		
Rank quintiles	No	No	No	Yes	No	Yes	No
Controls ^a	Yes	No	Yes	Yes	Yes	Yes	No
Competition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,872	1,872	3,368	3,368	5,021	5,021	1,843
R ²	0.47	0.42	0.35	0.35	0.27	0.27	0.42

Panel B							
Dependent variable:		VC_i^{post}					$VC Deals_i^{post}$
Sample:	All applicants		No prev. apps		>2 prev. wins		No prev. wins
	1	All	2	All	1	All	1
Bandwidth:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Award	0.11 (0.023)	0.075 (0.026)	0.11 (0.037)	0.10 (0.048)	0.080 (0.048)	0.074 (0.053)	0.93 (0.19)
Normalized rank, normalized rank ²	No	Yes	No	Yes	No	Yes	No
Competition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	No ^b
Observations	2,714	6,400	1,514	2,951	1,246	2,322	1,872
R ²	0.37	0.19	0.50	0.28	0.51	0.35	0.051 (Pseudo- R ²)

Notes: This table reports regression estimates of the Phase 1 award effect on subsequent VC investment using variants of equation (1). The model is OLS, except for column 7 in panel B, which is negative binomial. Panel A varies the bandwidth around the cutoff and controls in columns 1–6, and uses the amount of VC investment as dependent variable in 7. Panel B varies the sample (as in Table 2) in 1–6, and uses the number of VC deals as dependent variable in column 7.

^a Controls are VC_i^{prev} and previous non-DOE SBIR awards.

^b This model uses year sector fixed effects. Standard errors are clustered by sector year. Year > 1005.

Table 3: Phase 1 grant impact on VC

6.4 Figure 4 and Table 4: Revenue effects

Read:

- Figure 4 shows a discontinuity in log revenue around the award cutoff.
- Table 4 uses OLS, Tobit, and zero-inflated negative binomial models.
- OLS estimates in panel A imply revenue effects around \$1.3–\$1.7 million in the broader sample.
- Tobit estimates in panel B suggest larger effects, around \$2.3–\$2.5 million.
- The zero-inflated models show that the grant affects both the positive-revenue margin and the revenue distribution.

Economic message: Phase 1 grants do not only increase patents and investor interest. They also help firms reach commercialization and produce revenue.

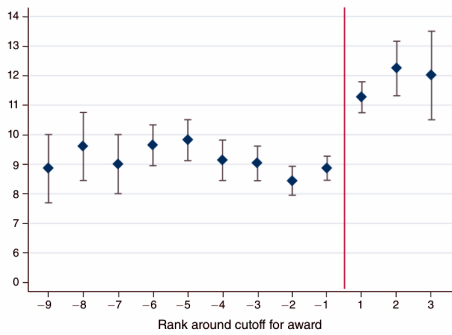


FIGURE 4. LOG REVENUE (in \$Millions)

Figure 4: Log revenue by rank

TABLE 4—PHASE 1 GRANT IMPACT ON REVENUE (in \$Millions)

Panel A							
Model:		OLS			Zero-inflated negative binomial (ZINB)		
Bandwidth:		1	All	2	1	All	All
		(1)	(2)	(3)	(4)	(5)	(6)
Award ^a		1.7 (0.93)	1.3 (0.58)	0.19 (0.11)	0.25 (0.12)	0.29 (0.13)	0.34 (0.096)
Award (logistic ZINB)					-0.67 (0.16)	-0.82 (0.17)	-0.76 (0.13)
Norm. rank, Norm. rank ²		No	No	No	No	Yes	No
Rank quintiles		No	Yes	No	No	No	Yes
Controls ^b		No	No	No	No	Yes	No
Competition fixed effects		Yes	Yes	Yes	No	No	No
Sector and year fixed effects		No	No	No	Yes	Yes	Yes
Observations		1,108	3,942	1,176	1,108	3,942	3,942
R ² /Dev.-based R ²		0.40	0.14	0.16	0.11	0.10	0.06

(Continued)

Table 4, Panel A: Revenue effects

TABLE 4—PHASE 1 GRANT IMPACT ON REVENUE (in \$millions) (Continued)

Panel B Model:	Tobit			Zero-inflated negative binomial (ZINB)		
	1	All	All	1	1	1
	No previous winners			All apps	No prev. apps	>2 prev. wins
Sample:	(1)	(2)	(3)	(4)	(5)	(6)
Award ^a	2.3 (0.72)	2.5 (0.72)	2.5 (0.59)	0.53 (0.10)	0.39 (0.16)	0.34 (0.26)
Award (logistic ZINB)				−0.91 (0.14)	−1.0 (0.20)	−1.0 (0.33)
Norm. rank, Norm. rank ²	No	Yes	No	No	No	No
Rank quintiles	No	No	Yes	No	No	No
Sector and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,108	3,942	3,942	1,384	657	329
Pseudo R^2 /Dev-based R^2	0.002	0.001	0.001	0.10	0.17	0.18

Notes: This table reports regression estimates of the Phase 1 award effect on revenue (in millions of nominal \$) using variants of equation (1). Panel A varies the bandwidth around the cutoff and uses OLS (columns 1–3) and negative binomial (columns 4–6) models. Panel B, columns 1–3 use Tobit models. Panel B, columns 4–6 vary the sample (as in Table 2).

^aControls are VC_i^{prev} and previous non-DOE SBIR awards. * Sample is restricted to firms with revenue > 0.

^bIn columns 4–6, this coefficient is on the full ZINB model. Sector and year fixed effects used where ML estima-

Table 4, Panel B: Revenue effects continued

6.5 Table 5 and Table 6: Survival, exit, and Phase 2 effects

Read:

- Table 5 shows that Phase 1 increases successful exit by about 4.4–5 percentage points and increases the probability that the firm remains in business by about 12–15 percentage points.
- Rank controls do not explain these survival and exit effects.
- Table 6 shows that Phase 2 awards have a small positive patent effect in some specifications but little measurable effect on VC, revenue, survival, or exit.
- The lack of Phase 2 effects contrasts sharply with Phase 1 because many high-quality Phase 1 winners that receive VC do not apply for Phase 2.

Economic message: Early financing affects the firm life cycle, but later-stage public funding is less transformative for marginal firms. The strongest policy effect is at the proof-of-concept stage.

TABLE 5—PHASE 1 GRANT IMPACT ON SURVIVAL AND SUCCESSFUL EXIT

Dependent variable:	$Exit_i^{post}$				$InxtrmBus_i^{post}$			
	1	All	All	All	1	All	All	All
Bandwidth:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Award	0.046 (0.025)	0.05 (0.022)	0.046 (0.022)	0.044 (0.016)	0.15 (0.061)	0.14 (0.046)	0.14 (0.048)	0.12 (0.034)
Normalized rank	−3.6e-4 (0.0047)	−9.1e-4 (0.0045)			−0.0027 (0.0093)	5.2e-4 (0.01)		
Normalized rank ²	5.9e-5 (2.9e-4)	1.1e-6 (2.8e-4)			3.6e-4 (4.5e-4)	3.1e-4 (5e-4)		
Rank quintiles	No	No	No	Yes	No	No	No	Yes
Controls ^a	No	Yes	No	No	No	Yes	No	No
Competition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,872	5,021	5,021	5,021	1,212	3,880	3,880	3,880
R^2	0.38	0.20	0.17	0.17	0.46	0.27	0.25	0.25

Notes: This table reports regression estimates of the Phase 1 award effect on survival (remaining in business, columns 1–4) and successful exits (IPOs and acquisitions, columns 5–8) using variants of equation (1).

^aControls are VC_i^{prev} and previous non-DOE SBIR awards. Standard errors are clustered by sector-year. Year

Table 5: Survival and successful exit

TABLE 6—IMPACT OF PHASE 2 GRANT

Panel A. Cite-weighted patents									
Model:	Negative binomial						OLS		
Dependent variable:	$Cites_i^{post}$						$\ln(1 + Cites_i^{post})$		
Sample:	No previous winners			All applicants		>1 prev. win	No previous winners		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Phase 2 award	0.77 (0.29)	0.69 (0.30)	0.34 (0.17)	0.28 (0.15)	0.25 (0.17)	0.17 (0.15)	0.29 (0.13)	0.32 (0.15)	0.22 (0.12)
Phase 1 award			0.33 (0.10)						0.17 (0.061)
Controls ^a	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes
Sector and year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	408	408	5,021	867	867	459	408	408	5,021
[Pseudo-] R^2	0.13	0.091	0.21	0.092	0.042	0.10	0.51	0.35	0.42

Panel B. VC, revenue, survival, and exit										
Model:	OLS					OLS	ZINB	OLS		
Dependent variable:	VC_i^{post}					$Revenue_i$	$InBus_i^{post}$	$Exit_i^{post}$		
Sample:	No previous winners			All apps		>1 prev. win	No previous winners			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
Phase 2 award	0.027 (0.036)	0.027 (0.040)	-0.016 (0.034)	0.049 (0.032)	0.057 (0.051)	-0.06 (0.21)	-0.25 (0.15)	0.050 (0.053)	-0.0025 (0.027)	
Phase 1 award			0.11 (0.021)							
Controls ^a	Yes	No	Yes	No	No	No	No	No	No	
Sector and year FE	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	
Observations	408	408	5,021	867	459	226	474	311	408	
Pseudo- R^2	0.25	0.14	0.13	0.080	0.12	0.16	0.004	0.19	0.13	

Notes: This table reports estimates of the Phase 2 grant effect on all outcomes, using variants of equation (1). The model varies based on the dependent variable. There is no rank control due to the small number of applicants in each specification.

Table 6: Phase 2 grant effects

6.6 Table 7 and Table 8: Heterogeneity and technology categories

Read:

- Table 7 shows that the grant effect is stronger for firms with no previous cite-weighted patents, for hardware firms, for young firms, and for firms in emerging sectors.
- The VC effect is especially large for hardware firms, consistent with greater capital needs and collateral constraints.
- Table 8 shows technology-by-technology effects.
- The largest VC effects appear in areas such as geothermal, hydropower/wave/tidal, solar, carbon capture/storage, and efficiency technologies.

Economic message: The grant is most valuable where financing frictions should bind hardest. This is the paper’s strongest heterogeneity evidence for the financing-constraint mechanism.

TABLE 7—HETEROGENEITY IN PHASE 1 GRANT IMPACT ACROSS FIRM CHARACTERISTICS

Dependent variable:	Zero versus ≥ 1 cite-weighted patents		Hardware versus software	
	VC_i^{post}	$lnBus_i^{post}$	VC_i^{post}	$lnBus_i^{post}$
Bandwidth:	3	3	2	2
	(1)	(2)	(3)	(4)
Award $\cdot (1 No Cites_i^{pre})$	0.057 (0.033)	0.31 (0.1)		
Award $\cdot (1 Hardware_i)$			0.32 (0.19)	-0.038 (0.055)
Award	0.1 (0.023)	-0.08 (0.089)	0.083 (0.14)	0.13 (0.055)
$1 No Cites_i^{pre}$	-0.11 (0.01)	0.23 (0.42)		
$1 Hardware_i$			0.0064 (0.18)	0.56 (0.021)
Sector-year fixed effect and normal rank	Yes	Yes	Yes	Yes
Observations	3,368	2,378	1,209	1,209
R^2	0.209	0.107	0.529	0.530

Dependent variable:	Age ≤ 2 yrs versus > 2 yrs		Emerging versus mature sector	
	VC_i^{post}	$lnBus_i^{post}$	VC_i^{post}	$lnBus_i^{post}$
Bandwidth:	3	3	3	3
	(5)	(6)	(7)	(8)
Award $\cdot (1 Age_i \leq 2)$	0.14 (0.074)	0.15 (0.085)		
Award $\cdot (1 Emerging Sector_i)$			0.24 (0.098)	0.29 (0.16)
Award	0.033 (0.027)	0.09 (0.054)	-0.041 (0.062)	0.039 (0.13)
$1 Age_i \leq 2$	0.26 (0.12)	-0.53 (0.18)		
$1 Emerging Sector_i$			-0.055 (0.063)	0.17 (0.1)
Sector-year fixed effect and normal rank	Yes	Yes	Yes	Yes
Observations	3,368	2,378	1,772	1,238
R^2	0.092	0.096	0.160	0.234

Notes: This table reports estimates of the difference by firm type in the Phase 1 grant effect on VC investment and survival using variants of equation (1) (e.g., coefficient on Award $\cdot (1 | Age_i \leq 2)$ means the grant effect is 14 pp higher for younger firms). All regressors, including fixed effects and rank, are interacted with the firm type indicator. Bank controls and sector year fixed effects are included. Standard errors robust and clustered at the sector year

Table 7: Heterogeneity in Phase 1 impact

TABLE 8—IMPACT OF PHASE 1 GRANT BY TECHNOLOGY (Coefficient on award shown)

	VC_i^{post}	$ln(1 + Cites_i^{post})$	$lnBus_i^{post}$	Observations
	(1)	(2)	(3)	
Geothermal	0.56 (0.24)	-0.58 (0.30)	0.14 (0.13)	51
Hydropower, wave and tidal	0.51 (0.19)	0.50 (0.20)	0.26 (0.025)	236
Solar	0.25 (0.11)	0.15 (0.13)	0.20 (0.077)	421
Carbon capture and storage	0.20 (0.091)	0.73 (0.34)	0.25 (0.13)	211
Building and lighting efficiency	0.14 (0.057)	0.39 (0.16)	0.21 (0.048)	370
Vehicles, motors, engines, batteries	0.12 (0.060)	0.23 (0.12)	0.16 (0.059)	726
Wind	0.11 (0.039)	0.030 (0.10)	0.15 (0.093)	194
Advanced materials	0.11 (0.071)	-0.081 (0.18)	0.23 (0.077)	435
Biomass production/generation	0.085 (0.067)	0.036 (0.29)	0.21 (0.079)	308
Fuel cells and hydrogen	0.077 (0.072)	0.17 (0.18)	0.27 (0.10)	400
Natural gas	0.060 (0.074)	0.54 (0.45)	-0.15 (0.17)	255
Recycling, waste-to-energy, and water	0.045 (0.053)	0.094 (0.21)	0.061 (0.08)	549
Smart grid, sensors, converters	0.045 (0.053)	0.36 (0.21)	0.056 (0.069)	634
Air and emission control	0.025 (0.035)	0.20 (0.24)	0.20 (0.11)	300
Coal	0.024 (0.053)	0.79 (0.51)	0.11 (0.41)	108
Biofuels and biochemicals	0.014 (0.054)	0.036 (0.29)	0.051 (0.13)	176

Notes: This table reports regression estimates of the Phase 1 grant effect on VC investment, log cite-weighted patents, and survival by technology type using variants of equation (1). Each row represents a separate regression whose sample is limited to competitions in the technology type. Only the coefficient on treatment is shown. All models use OLS and use all the data. Other and OLS are omitted due to few observations. Standard errors robust and

Table 8: Impact by technology

6.7 Figure 5 and Table 9: Timing and grant use

Read:

- Figure 5 shows that a large part of the effect on VC and cite-weighted patents appears within the first several years after the grant.
- VC effects occur quickly, consistent with the grant helping firms become investable.
- Patent effects accumulate over time but much of the effect appears early.
- Table 9 shows survey evidence on how firms used grants.
- Phase 1 recipients report using funds mainly for basic R&D and testing/demonstrating/prototyping.

Economic message: The timing and survey evidence support the prototyping channel. The money is used to reduce technological uncertainty and make the firm more credible to investors.

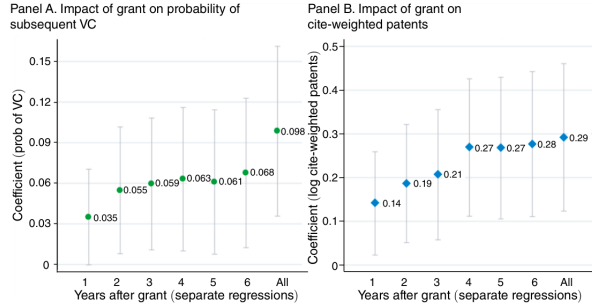


FIGURE 5. PHASE 1 GRANT IMPACT ON VC AND CITE-WEIGHTED PATENTS OVER TIME

Figure 5: Effects over time

6.8 Table 10 and Table 11: Robustness and placebo tests

Read:

- Table 10 tests whether the effect is really driven by winning ranks across competitions with different numbers of awards.
- Positive and meaningful effects are concentrated on winning ranks rather than arbitrary raw ranks.
- Table 11 uses artificial cutoffs among losers or among lower-ranked firms.
- Placebo awards around artificial cutoffs do not reproduce the main treatment effects.

Economic message: The robustness tests reinforce the RD interpretation: the estimated effects arise from receiving the grant, not from rank trends or arbitrary discontinuities.

TABLE 10—IDENTIFYING EFFECT WITH NUMBER OF AWARDS IN COMPETITION

Dependent variable: VC_i^{post}	(1)	(1) Continued
$(R_i^{Raw} = 1) \cdot (1 \#Awards = 1)$	0.093 (0.042)	$(R_i^{Raw} = 1) \cdot (1 \#Awards = 3)$ 0.095 (0.087)
$(R_i^{Raw} = 2) \cdot (1 \#Awards = 1)$	-0.055 (0.093)	$(R_i^{Raw} = 2) \cdot (1 \#Awards = 3)$ 0.16 (0.12)
$(R_i^{Raw} = 3) \cdot (1 \#Awards = 1)$	0.060 (0.036)	$(R_i^{Raw} = 3) \cdot (1 \#Awards = 3)$ 0.055 (0.090)
$(R_i^{Raw} = 4) \cdot (1 \#Awards = 1)$	0.015 (0.034)	$(R_i^{Raw} = 4) \cdot (1 \#Awards = 3)$ -0.12 (0.10)
$(R_i^{Raw} = 1) \cdot (1 \#Awards = 2)$	0.13 (0.063)	$(R_i^{Raw} = 1) \cdot (1 \#Awards = 4)$ 0.087 (0.16)
$(R_i^{Raw} = 2) \cdot (1 \#Awards = 2)$	0.15 (0.081)	$(R_i^{Raw} = 2) \cdot (1 \#Awards = 4)$ 0.18 (0.22)
$(R_i^{Raw} = 3) \cdot (1 \#Awards = 2)$	0.064 (0.065)	$(R_i^{Raw} = 3) \cdot (1 \#Awards = 4)$ -0.017 (0.17)
$(R_i^{Raw} = 4) \cdot (1 \#Awards = 2)$	0.027 (0.049)	$(R_i^{Raw} = 4) \cdot (1 \#Awards = 4)$ 0.14 (0.18)
Observations		3,206
R^2		0.29

Notes: This table reports regression estimates of the Phase 1 grant effect on subsequent VC investment in which identifying variation comes from having a given raw rank in a competition in which the number of awards is at least that rank. Raw rank dummies are interacted with dummies for the competition's number of awards. The omitted dummy in each group is $(R_i^{Raw} = 5) \cdot (1 | \#Awards = x)$. Only competitions with 1–4 awards and firms with raw ranks of 1–5 included. Standard errors robust and clustered at sector-year level. Year ≥ 1995 .

Table 10: Effect by number of awards

TABLE 9—GRANT USE SURVEY RESULTS

Survey response:	Fraction			Frequency		
	Yes (1)	No (2)	If random (3)	Yes (4)	No (5)	χ^2 p-value ^a (6)
Panel A. Phase 1 (94 respondents)						
(a) Basic R&D on new technology	0.71	0.29	0.40	67	27	0.00
(b) Patent applications	0.08	0.92	0.40	8	86	0.00
(c) Testing existing technology	0.65	0.35	0.40	61	33	0.00
(d) Overhead/salaries	0.47	0.53	0.40	44	50	0.18
(e) Something else	0.10	0.90	0.40	9	85	0.00
Only (a)	0.32	0.68	0.088	30	64	0.00
Only (c)	0.26	0.74	0.088	24	70	0.00
Neither (a) nor (c)	0.032	0.97	0.36	3	91	0.00
Panel B. Phase 2 (38 respondents)						
(a) Basic R&D on new technology	0.42	0.58	0.50	16	22	0.33
(b) Patent applications	0.21	0.79	0.50	8	30	0.00
(c) Testing existing technology	0.79	0.21	0.50	30	8	0.00
(d) Overhead/salaries	0.63	0.37	0.50	24	14	0.11
(e) Something else	0.18	0.82	0.50	7	31	0.00
Only (a)	0.026	0.97	0.042	1	37	0.00
Only (c)	0.16	0.84	0.042	6	32	0.00
Neither (a) nor (c)	0.11	0.89	0.25	4	34	0.04

Notes: This table shows the fraction (columns 1–2) and frequency (columns 4–5) of responses to the grant use survey. For example, the first row of panel A indicates that 71 percent of respondents reported using the grant for basic R&D.

^a χ^2 goodness-of-fit test: asks whether observed proportion differs from random assignment. The null for Phase 1 is that 40 percent pick any given option (respondents averaged 2 choices). If the observed fraction is different from 40 percent with 95 percent confidence, then p-value < 0.05. For Phase 2, the null is that 50 percent pick an option

Table 9: Grant use survey results

	VC_i^{post} (1)	$\ln(1 + Cites_i^{post})$ (2)	Revenue _i (3)	$\ln Bus_i^{post}$ (4)	Exit _i ^{post} (5)
Panel A. Artificial cutoff between ranks 1 and 2					
Award	-0.19 (0.19)	0.044 (0.55)	-3.0 (0.61)	0.0 (0.23)	-0.021 (0.1)
Competition fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	599	599	420	450	599
R^2	0.78	0.80	0.80	0.85	0.75
Panel B. Artificial cutoff between ranks -1 and -2					
Award	0.023 (0.023)	0.26 (0.1)	-0.32 (0.53)	0.074 (0.052)	0.0033 (0.014)
Competition fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	2,261	2,261	1,370	1,486	2,261
R^2	0.36	0.41	0.41	0.39	0.341

Notes: This table reports regression estimates of the placebo Phase 1 grant effect on all the outcomes using a OLS model and a bandwidth of 1 firm around the cutoff. The regressions are variants of equation (1), except the cutoff for Award is moved from between ranks -1 and 1 to between 1 and 2 (panel A) or -1 and -2 (panel B). Standard errors are robust and clustered at sector-year level. Year ≥ 1995 .

Table 11: Placebo tests

7 Short summary

- The paper evaluates DOE SBIR R&D grants using ranked application data and a regression discontinuity design.
- Phase 1 grants are small early-stage grants around the proof-of-concept stage.
- Phase 1 increases cite-weighted patents, venture capital, revenue, survival, and successful exit.
- The VC effect is especially important: the grant approximately doubles the probability of receiving VC.

- Phase 2 grants have much weaker effects.
- Heterogeneity results show stronger effects for firms that are more financially constrained.
- Certification alone is unlikely to explain the results because DOE ranks are not strongly predictive of market outcomes.
- Survey evidence shows that firms use grants primarily for R&D and prototyping.
- The paper’s main lesson is that small early-stage public grants can catalyze private capital and commercialization.

8 Conclusion

8.1 Core conclusion

Howell shows that early-stage R&D grants can have large causal effects on startup outcomes. Phase 1 SBIR awards increase innovation, private financing, revenue, survival, and successful exit for firms close to the award cutoff. The evidence indicates that grants alleviate financing frictions rather than crowd out private capital.

8.2 Economic intuition

Young innovative firms often have promising but unproven technologies. Private investors face uncertainty, the assets are intangible, and founders may not be able to finance proof-of-concept work. A small grant lets the firm build, test, and demonstrate the technology. Once uncertainty falls, outside investors become more willing to provide capital.

8.3 Incremental contribution in one sentence

The paper provides the first large-sample quasi-experimental evidence that early government R&D grants can causally transform financially constrained startups into patenting, VC-backed, revenue-generating firms.

8.4 Takeaway

The policy lesson is not that all public R&D subsidies work equally well. The evidence points to a more specific design principle: small, early-stage grants targeted to firms facing severe financing constraints can have high leverage because they enable private follow-on financing and commercialization.